Jônatas Augusto Manzolli

Postdoctoral Researcher | Electric Mobility, Optimization, Energy Systems, and AI for Transportation

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Short Bio

Postdoctoral researcher and electrical engineer working at the intersection of electric mobility, transportation systems, optimization, artificial intelligence, and energy-system integration. Research focuses on electric bus fleet planning and operations, smart charging, V2G/grid services, battery degradation, agent-based simulation, robust and hierarchical optimization, and multi-criteria decision analysis for transport policy. Publication record includes *Applied Energy*, *Energy*, *Renewable and Sustainable Energy Reviews*, *eTransportation*, *Sustainable Futures*, and *npj Sustainable Mobility and Transport*.

Areas of Expertise

Electric mobility	Electric bus fleet planning and operations; EV adoption; charging infrastructure; fleet electrification strategy.
Optimization	Robust, bi-level, hierarchical, and multi-objective optimization; MILP/MINLP; metaheuristics.
Energy systems	V2G; energy markets; tariff design; distributed energy resources; grid-aware fleet operation.
AI and simulation	Agent-based simulation; machine learning for energy consumption and battery degradation; digital twins.
Decision support	MCDA; SMAA; PROMETHEE; participatory and synthetic MCDA; policy-oriented transportation planning.

Education

University of Coimbra, Portugal — Sept. 2020–Feb. 2025
Ph.D. in Sustainable Energy Systems
Thesis: *Adaptive Energy Management Strategies for Electric Bus Fleets: A Hybrid Approach with Artificial Intelligence and Optimisation Methods.* Summa cum laude.

University of Coimbra, Portugal — Sept. 2018–Mar. 2020
M.Sc. in Energy for Sustainability
Dissertation: *Decision Support for Planning a Bus Rapid Transit Charging Infrastructure.*

Technical University of Munich, Germany — Mar. 2015–Mar. 2016
Exchange Program in Electrical Engineering

University of Campinas, Brazil — Mar. 2011–Mar. 2017
B.Sc. in Electrical Engineering
Best undergraduate final project award.

Working Experience

McGill University, IMaTS Lab — May 2025–Present
Postdoctoral Researcher

- Conduct research on electric mobility, autonomous and shared mobility systems, battery degradation prediction, and AI-enabled transportation-energy integration.
- Participated in McGill University's Innovation to Impact program and Lab2Market Validate commercialization activities.

- Delivered lectures and seminars on optimization, electric mobility, and integrated transport-energy systems.

INESC Coimbra / University of Coimbra June 2019–Present

Affiliated Researcher; Researcher and Optimization Specialist

- Developed optimization, simulation, and decision-support frameworks for urban electric bus fleets, charging infrastructure, V2G, energy-market participation, and uncertainty-aware operations.
- Built robust, hierarchical, and multi-criteria models to support electric bus fleet planning and operations under operational, weather, and battery-aging uncertainty.
- Implemented models and data workflows in Python and MATLAB, using operations research tools including Pyomo, Gurobi, CPLEX, GAMS, SciPy, and JSMAA.

University of Sherbrooke, Canada Oct. 2022–Mar. 2023

Visiting Researcher

- Developed semi-empirical models to predict electric bus battery aging under temperature and operating-condition variability.
- Conducted laboratory tests and analyzed battery data to improve model accuracy, reliability, and practical applicability.

NOVA Cidade – Urban Analytics Lab, Lisbon, Portugal Aug. 2020–Oct. 2020

Student Researcher, C-Tech Project (MIT Portugal)

- Developed an MCDA framework to map street-level walkability in Lisbon, integrating operations research with urban analytics.

IAV GmbH, Munich, Germany Sept. 2015–Mar. 2016

Intern

- Worked on BMW engine design projects, embedded monitoring systems, component modeling, simulation, and data visualization.

Awards and Distinctions

Year	Recognition	Project / Context
2025	Best Paper Award	Transportation Research Board Annual Meeting, Standing Committee on Critical Infrastructure Protection.
2021	Altice Innovation Award	Second place, Lisbon, Portugal; innovation prize connected to sustainable technology and entrepreneurship.
2021	CityHack Hackathon	First place for ReneWaste, a digital platform connecting waste-management companies and businesses.
2021	Shift APPENS Hackathon	First place for ReneWaste.
2020	Microsoft Building the Future Hackathon	Second place for Blockcharging.
2020	Oeiras Valley Competition	Top 10 for a fast-charging allocation strategy.
2020	EDP University Challenge	Top 10 for Go! Charging!
2017	Undergraduate distinction	Best undergraduate final project award, University of Campinas.

Funding and Fellowships

Year	Funding / Fellowship	Type	Amount (EUR equiv.)
2025	SYLFF Research Grant	Research grant	EUR 13,800
2025	Lab2Market Validate Summer Cohort	Commercialization funding	EUR 6,800
2023	Tokyo Foundation SYLFF Research Grant	Research grant	EUR 2,700
2022	GISU International Urban Entrepreneurship, Padova, Italy	Entrepreneurship prize	EUR 10,000
2021	Foundation for Science and Technology (FCT), National PhD Research Fellowship	Doctoral fellowship	EUR 68,000
2020	Sasakawa Young Leaders Fellowship Fund (SYLFF) Fellowship Program	Fellowship	EUR 10,200
2014	Science without Borders Scholarship, DAAD/CAPES	International scholarship	EUR 20,400

Publications

Peer-reviewed journal articles

1. **J. A. Manzolli**, A. V. D'Apice, P. K. Pandey, F. Ciari, and L. Miranda-Moreno, "Planning resilient electric bus operations in cold regions: An agent-based simulation-optimization framework", *Applied Energy*, vol. 413, 127735, 2026. doi: 10.1016/j.apenergy.2026.127735.
2. **J. A. Manzolli**, J. Yu, A. V. D'Apice, and L. Miranda-Moreno, "Balancing energy resilience and mobility: A multi-objective strategy for deploying shared autonomous electric vehicles during power outages", *npj Sustainable Mobility and Transport*, vol. 3, no. 1, article 13, 2026. doi: 10.1038/s44333-026-00081-9.
3. D. Deda, **J. A. Manzolli**, M. J. Quina, and H. Gervasio, "The Road to 2030: Combining Life Cycle Assessment and Multi-Criteria Decision Analysis to Evaluate Commuting Alternatives in a University Context", *Sustainability*, vol. 17, no. 13, 5839, 2025. doi: 10.3390/su17135839.
4. **J. A. Manzolli**, J. Yu, and L. Miranda-Moreno, "Synthetic multi-criteria decision analysis (S-MCDA): A new framework for participatory transportation planning", *Transportation Research Interdisciplinary Perspectives*, vol. 31, 101463, 2025. doi: 10.1016/j.trip.2025.101463.
5. **J. A. Manzolli**, P. Messier, J. P. Trovao, and C. H. Antunes, "Decision-making in bus-transit systems: A comprehensive approach based on stochastic multi-criteria acceptability analysis", *Sustainable Futures*, vol. 9, 100653, 2025. doi: 10.1016/j.sftr.2025.100653.
6. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "Electric bus fleet charging management: A robust optimisation framework addressing battery ageing, time-of-use tariffs, and energy consumption uncertainty", *Applied Energy*, vol. 381, 125137, 2025. doi: 10.1016/j.apenergy.2024.125137.
7. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "Aggregator-supported strategy for electric bus fleet charging: A hierarchical optimisation approach", *Energy*, vol. 306, 132497, 2024. doi: 10.1016/j.energy.2024.132497.
8. I. Cavalcante, J. Junior, **J. A. Manzolli**, L. Almeida, M. Pungo, C. P. Guzman, and H. Morais, "Electric Vehicles Charging Using Photovoltaic Energy Surplus: A Framework Based on Blockchain", *Energies*, vol. 16, no. 6, 2694, 2023. doi: 10.3390/en16062694.
9. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "Electric bus coordinated charging strategy considering V2G and battery degradation", *Energy*, vol. 254, 124252, 2022. doi: 10.1016/j.energy.2022.124252.
10. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "A review of electric bus vehicles research topics – Methods and trends", *Renewable and Sustainable Energy Reviews*, vol. 159, 112211, 2022. doi: 10.1016/j.rser.2022.112211.
11. **J. A. Manzolli**, A. Oliveira, and M. de Castro Neto, "Evaluating Walkability through a Multi-Criteria Decision Analysis Approach: A Lisbon Case Study", *Sustainability*, vol. 13, no. 3, 1450, 2021. doi: 10.3390/su13031450.
12. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "Scenario-Based Multi-criteria decision analysis for rapid transit systems implementation in an urban context", *eTransportation*, vol. 7, 100101, 2021. doi: 10.1016/j.etran.2020.100101.

Book chapters

1. R. D. de Castro, **J. A. Manzolli**, N. C. Figueiredo, and P. P. da Silva, "Renewable Power Purchase Agreements: An Instrument for Corporate Decarbonization", in *Wiring the Future: Financial Strategies, Challenges, and Opportunities in the Sustainable Energy Transition*, Springer Nature Switzerland, 2026, pp. 13–41. doi: 10.1007/978-3-032-14717-2_2.
2. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "Forecasting of Vehicle Electrification in Modern Power Grids", in *Vehicle Electrification in Modern Power Grids: Innovative Perspectives on Power Electronics Technologies and Control Challenges*, Elsevier, 2024. doi: 10.1016/B978-0-443-13969-7.00003-5.
3. D. Deda, **J. A. Manzolli**, J. A. Santos, W. V. D. S. Correia, P. M. A. R. Carvalho, F. B. Carvalho, M. B. Carvalho, et al., "Transicao e Pobreza Energetica", in *Desafios Societais e a Investigacao em Direito*, vol. 6, 2023. doi: 10.47907/DesafiosSocietais6/2024.

Conference papers, posters, and proceedings

1. R. D. de Castro, **J. A. Manzolli**, J. P. Trovao, C. H. Antunes, N. C. Figueiredo, and P. P. da Silva, "Strategic Energy Procurement for EV Fleets: A Comparative Study", 2025 21st International Conference on the European Energy Market (EEM), 2025. doi: 10.1109/EEM64765.2025.11050169.
2. **J. A. Manzolli**, W. Do, C. H. Antunes, J. P. Trovao, and L. Miranda-Moreno, "An Agent-based Optimization Framework for Smart Electric Bus Charging with Operational Strategies", Transportation Research Board Annual Meeting, Washington, DC, 2025. doi: 10.13140/RG.2.2.17321.30568.
3. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "An Integrated Optimization-MCDA Framework for Efficient Electric Bus System Planning," conference paper, 2024.
4. C. Hernandez, **J. A. Manzolli**, T. R. Simoes, D. J. Redol, D. C. Rodrigues, and D. F. Freire, "The Influence of a Rooftop Photovoltaic System on the Electricity Consumption of a Plastic Moulding Plant: A Carbon Footprint Assessment", 2024 IEEE Mediterranean Electrotechnical Conference (MELECON), 2024. doi: 10.1109/MELECON56669.2024.10608704.
5. J. Almeida, **J. A. Manzolli**, J. Soares, J. P. Trovao, F. Lezama, and C. H. Antunes, "Addressing Complexities in Electric Bus Fleet Charging: A Metaheuristic Approach," ELECTRIMACS 2024, Castello de la Plana, Spain, 2024.
6. **J. A. Manzolli**, W. Do, L. Miranda-Moreno, J. P. Trovao, and C. H. Antunes, "Optimising Electric Bus Fleet Charging Using a Simulation-Based Energy Consumption Model", 2023 IEEE Vehicle Power and Propulsion Conference (VPPC), 2023, pp. 1–6. doi: 10.1109/VPPC60535.2023.10403253.
7. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "Optimising Electric Bus Charging: Dynamic Tariffs in a Bi-Level Framework Considering Weather Conditions and Energy Storage", 21st IMACS World Congress, Rome, Italy, 2023. doi: 10.13140/RG.2.2.20170.13769.
8. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "Optimisation of an Electric Bus Charging Strategy Considering a Semi-Empirical Battery Degradation Model and Weather Conditions", 2022 11th International Conference on Control, Automation, and Information Sciences (ICCAIS), Hanoi, Vietnam, 2022, pp. 298–303. doi: 10.1109/ICCAIS56082.2022.9990180.
9. **J. A. Manzolli**, J. P. Trovao, and C. H. Antunes, "Electric Bus Smart Charging under a Bi-Level Optimisation Model to Set Dynamic Tariffs", IECON 2022 – 48th Annual Conference of the IEEE Industrial Electronics Society, 2022, pp. 1–6. doi: 10.1109/IECON49645.2022.9969101.

Selected Presentations, Tutorials, and Seminars

May 2026	**Electric Mobility and Integrated Optimization for Energy and Transport Systems**. Presentation, Climate, Emissions, and Energy in Transportation, McGill University, Montreal, Canada.
Apr. 2026	**Planning Resilient Electric Bus Operations**. Invited webinar, PANAMSTR Webinar 3.03: Optimization of Electric Vehicles, online.
Oct. 2025	**Scenario-Driven Optimization for Quebec's Electric Bus Fleets under Climate and Operational Variability**. Presentation and co-organized workshop, McGill University.
2025	**A Comprehensive Framework for Evaluating Different Transit Systems**. Congreso de Transporte y Logistica MexCan, Mexico City, Mexico.
2025	**Spatiotemporal Dynamics of Electric Vehicle Adoption in Quebec**. International Symposium on Transportation Data & Modelling, HEC Montreal.
Sept. 2025	**Nextdriv Platform Development and Electric Fleet Decision Support**. McGill Engine Startup Interns / Annual Engine Internship Showcase.
Oct. 2024	**Optimizing Charging Strategies for Electric Bus Fleets: A Robust Modelling Approach**. Tutorial, IEEE Vehicle Power and Propulsion Conference, Washington, DC.
May 2024	**Agent-Based Optimization for Electric Bus Transit Charging and Operation Strategies**. Seminar, Advances in Transit and Shared Mobility, McGill University.
May 2024	**Science Mapping Tools for Data Visualization and Literature Review**. Workshop, INESC Coimbra.
Dec. 2023	**Artificial Intelligence Tools for Transportation**. Seminar, University of Sherbrooke.
Sept. 2023	**Sustainable Transportation and Electric Vehicles**. Seminar, Polytechnic University of Coimbra.
Sept. 2023	**A Journey to the Future: Urban (R)evolution with Electric Bus Intelligence**. Seminar, McGill University.
Sept. 2023	**Energy Transition: Decarbonization in the Fuel Sector**. Round table, Greenfest, NOVA University, Lisbon.
June 2023	**The Path to a Net-Zero University: Coimbra as a Case Study**. Seminar, EU Sustainable Energy Week, Coimbra.
May 2023	**The Future of Transportation**. Talk, Pint of Science Portugal.
Apr. 2022	**Circular Ideas: How to Start a Sustainable Business?**. Workshop, Casa do Impacto, Lisbon.

Technical Skills

Programming	Python, MATLAB, C, HTML, CSS.
Optimization tools	Pyomo, PyRo, Gurobi, CPLEX, GAMS, SciPy, NEOS Server, JSMAA, metaheuristic implementations.
Data analysis	NumPy, Pandas, Matplotlib, statistical analysis, data cleaning, visualization, and reproducible workflows.
Science mapping	VOSviewer, Bibliometrix, systematic literature review workflows.
Design and front end	Figma, HTML, CSS, Streamlit, basic web prototyping.
Languages	Portuguese (native), English (fluent), German (advanced), Spanish (intermediate), Italian (basic).

Entrepreneurship-Related Activities

Ventures and technology transfer

Period	Initiative	Role and focus
2025–Present	Nextdriv	Founder / research commercialization initiative developing digital decision-support tools for electric fleet planning, operations, charging infrastructure, battery degradation, and energy-system analysis.
2021–2023	ReneWaste	Co-founder of a digital platform designed to increase urban recycling by connecting waste-management companies and businesses.

Competitions and hackathons

Year	Program	Result	Project
2021	CityHack Hackathon	First place	ReneWaste.
2021	Shift APPENS Hackathon	First place	ReneWaste.
2020	Microsoft Building the Future Hackathon	Second place	Blockcharging.
2020	Oeiras Valley Competition	Top 10	Fast-charging allocation strategy.
2020	EDP University Challenge	Top 10	Go! Charging!

Acceleration programs and training

Year	Program	Location / Organizer
2025	Lab2Market Validate	Canada; research commercialization and stakeholder discovery.
2024	Engine Innovation to Impact	McGill University, Montreal, Canada.
2022	European Innovation Academy	Porto, Portugal.
2022	International Entrepreneurial Week	EC2U Alliance, Pavia, Italy.
2022	Explorer Program	Santander X, Madrid, Spain.
2022	Triggers	Casa do Impacto, Lisbon, Portugal.
2019	International Workshop on Innovating	MIT, Boston, USA.

News, Media Coverage, and Public Outreach

Interviews and opinion pieces

Outlet	Year	Title / Theme
RTP	2023	"90 Segundos de Ciencia" research interview.
Sylff Association	2023	"Powering Up: How Electric Buses are Paving the Way for a Greener Tomorrow."
BRAS Center	2023	"Beyond a Car-Centric World."
Ambiente Magazine	2022	"Electric Mobility: Myths and the Importance of Demystifying Concepts."
Publico	2021	"Sustainable World: Coimbra Students Promote Webinar Week."

News articles

Outlet	Year	Article
Forbes	2025	"Why Smart Charging Is Key To Successful Electric Bus Fleet Operations."
INESC	2022	"INESC Coimbra Researcher Wins 2nd Place at the Hackathon Building the Future."
Noticias UC	2022	"UC Scientists Develop Intelligent Charging Model for Electric Bus Fleets."
Observador	2022	"University of Coimbra Develops Intelligent Charging Model for Electric Buses."
Diario de Noticias	2022	"University of Coimbra's Intelligent Charging Model for Electric Buses."
RTP	2022	"University of Coimbra's Intelligent Charging Model for Electric Buses."
Publico	2022	"University of Coimbra's Intelligent Charging Model for Electric Buses."
Portugal Resident	2022	"Coimbra Scientists Develop Intelligent Charging Model for Electric Buses."
FCTUC	2022	"UC Scientists Develop Intelligent Charging Model for Electric Bus Fleets."
Revista Sustentavel	2022	"Portuguese Develop Intelligent Charging Model that Reduces Costs and Increases Battery Life."
Ambiente Magazine	2022	"UC Scientists Create Intelligent Charging Model for Electric Bus Fleets."
Noticias de Coimbra	2022	"UC Develops Intelligent Charging Model for Electric Buses."

Selected Certificates

Area	Date	Certificate and competencies
Entrepreneurship	June 2019	Entrepreneurship and Innovation, MIT. Competencies: entrepreneurship, innovation, communication, leadership.
Entrepreneurship	May 2022	Entrepreneurism and Social Impact, Santander X. Competencies: entrepreneurship, innovation, communication, leadership.
Programming and AI	Apr. 2020	Python Programming, University of Michigan. Competencies: Python and programming.
Programming and AI	May 2020	Machine Learning, Stanford University. Competencies: programming, MATLAB, machine learning.
Programming and AI	June 2020	SPSS Advanced Course, APEU-FEUC / University of Coimbra. Competencies: statistics and mathematics.
Programming and AI	Feb. 2022	Neural Networks and Deep Learning, DeepLearning.AI. Competencies: Python, programming, machine learning.
Programming and AI	Dec. 2022	HTML and CSS in Web Projects, Alura. Competencies: CSS and HTML.
Programming and AI	May 2024	Energy Data Analytics, University of Texas at Austin. Competencies: Python, programming, data visualization.
Science communication	Nov. 2020	Public Speaking, Board of European Students of Technology (BEST).
Science communication	June 2021	Writing in the Sciences, Stanford University. Competencies: scientific writing.
Science communication	Mar. 2022	Science Communication and Media Training, Institute for Interdisciplinary Research, University of Coimbra.
Science communication	Jan. 2024	Creating Successful Research Posters, Springer Nature. Competencies: research, scientific communication, scientific posters.